2520090143

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CNNP – Skill 2

Stage 1:

```
kavya_aluru@Kavya:~$ sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd

Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
183 packages can be upgraded. Run 'apt list --upgradable' to see them.
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
iproute2 set to manually installed.
netcat-openbsd is already the newest version (1.234-1).
netcat-openbsd set to manually installed.
Upgrading:
  libpcre2-8-0  zlib1g

Installing:
  iperf3  mtr-tiny  tshark  wireshark

Installing dependencies:
  gnome-themes-extra-data  libqt6qmlworkerscript6
  ibverbs-providers        libqt6quick6
  libaacs0                 libqt6svg6
  libabsl20260107          libqt6waylandclient6
  libavcodec62             libqt6waylandcompositor6
  libavformat62            libqt6widgets6
  libavutil60              libqt6wlshellintegration6
  libb2-1                  librabbitmq4
  libbcg729-0              librav1e0.8
  libbdplus0               libre2-11
  libbluray3               librist4
  libcares2                libsbc1
  libchromaprint1          libsctp1
  libcjson1                libshine3
  libcodec2-1.2            libsmi2t64
  libdav1d7                libsnappy1v5
  libdouble-conversion3    libsoxr0
  libdvdnav4               libspandsp2t64
  libdvdread8t64           libspeexdsp1
  libevdev2                libsrt1.5-gnutls
  libfalcosecurity0t64     libssh-4
  libgme0                  libsvtav1enc2
```

Stage 2:

```
kavya_aluru@Kavya:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
kavya_aluru@Kavya:~$ sudo ip netns add router
Cannot create namespace file "/run/netns/router": File exists
kavya_aluru@Kavya:~$ sudo ip netns add host2
Cannot create namespace file "/run/netns/host2": File exists
kavya_aluru@Kavya:~$
```

```
kavya_aluru@Kavya:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
kavya_aluru@Kavya:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
```

```
kavya_aluru@Kavya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev ve
th-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
kavya_aluru@Kavya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev v
eth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
kavya_aluru@Kavya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev ve
th-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
kavya_aluru@Kavya:~$
```

Stage 4:

```
kavya_aluru@Kavya:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward
=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
```

Stage 5:

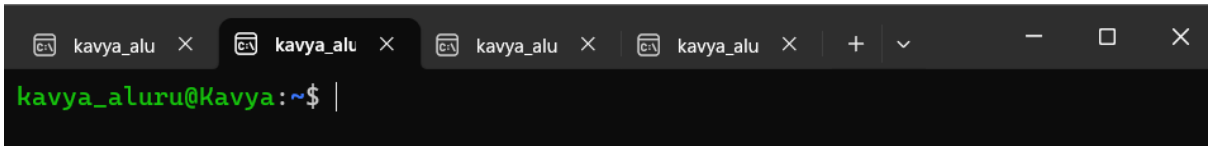
```
kavya_aluru@Kavya:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.054 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.046 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.054 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2055ms
rtt min/avg/max/mdev = 0.046/0.051/0.054/0.003 ms
kavya_aluru@Kavya:~$
```

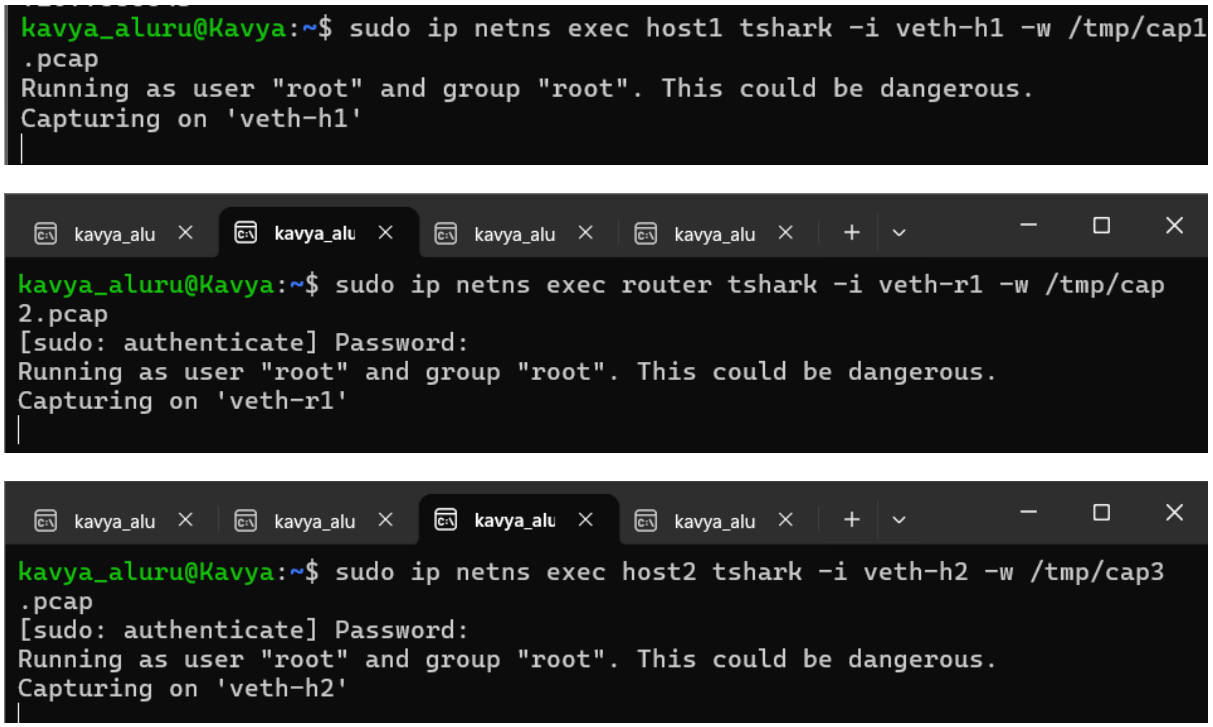
Stage 6:

```
kavya_aluru@Kavya:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root
 netem delay 20ms loss 1%
kavya_aluru@Kavya:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 42101455
72077885543
kavya_aluru@Kavya:~$
```

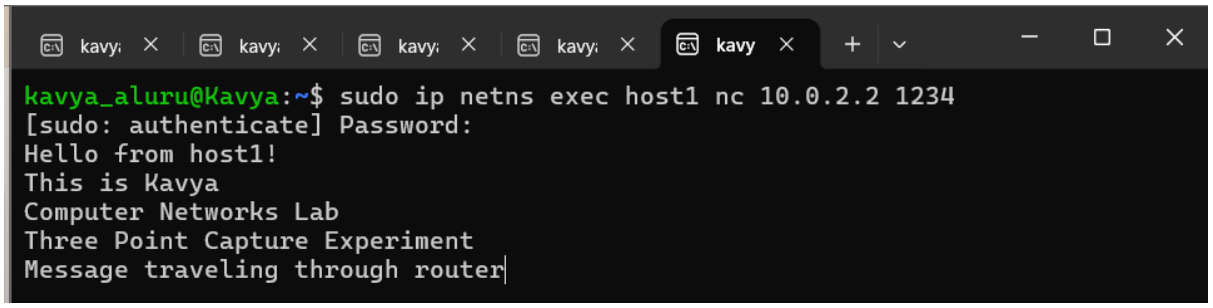
Stage 7:



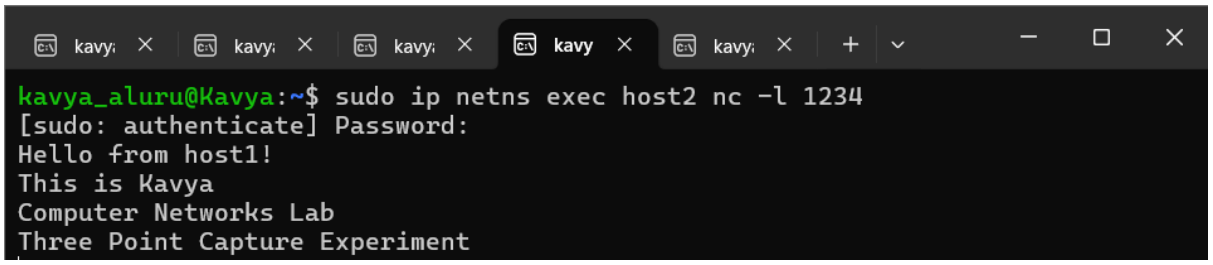
Stage 8:



Stage 9:



Stage 10:



Stage 11:

```
kavya_aluru@Kavya:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.
pcap
kavya_aluru@Kavya:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>
/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | un
iq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | un
iq -c
      5 63
      6 64
      5 63
      6 64
      6 63
      5 64
kavya_aluru@Kavya:~$
```

Stage 12:

```
kavya_aluru@Kavya:~$ sudo tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234 && t
cp.len > 0" -T fields -e tcp.payload | tr -d ':\n' | xxd -r -p
Running as user "root" and group "root". This could be dangerous.
Hello from host1!
This is Kavya
Three Point Capture Lab
Message travelling through router
kavya_aluru@Kavya:~$
```

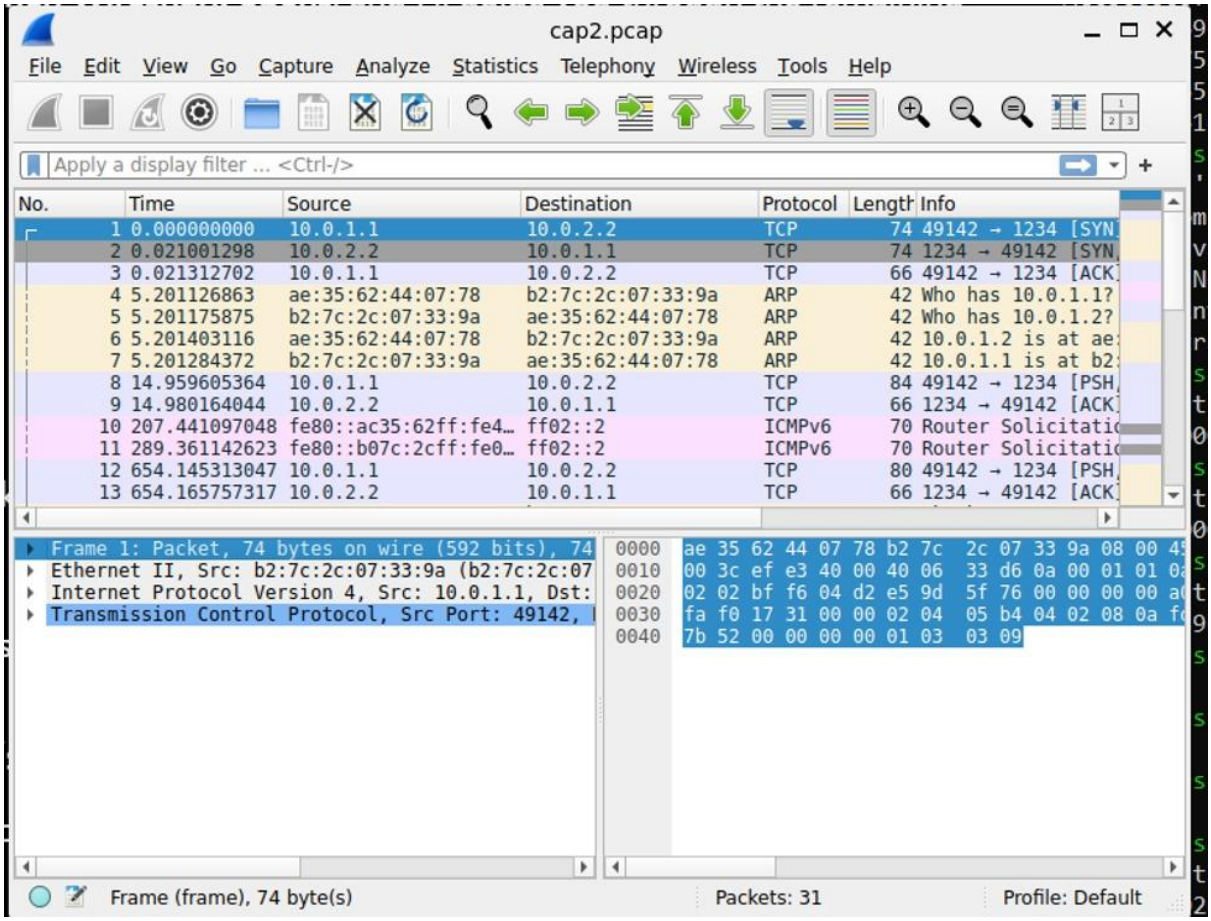
```
kavya_aluru@Kavya:~$ sudo apt install -y tshark iproute2 netcat-openbsd wire
shark
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 181
```

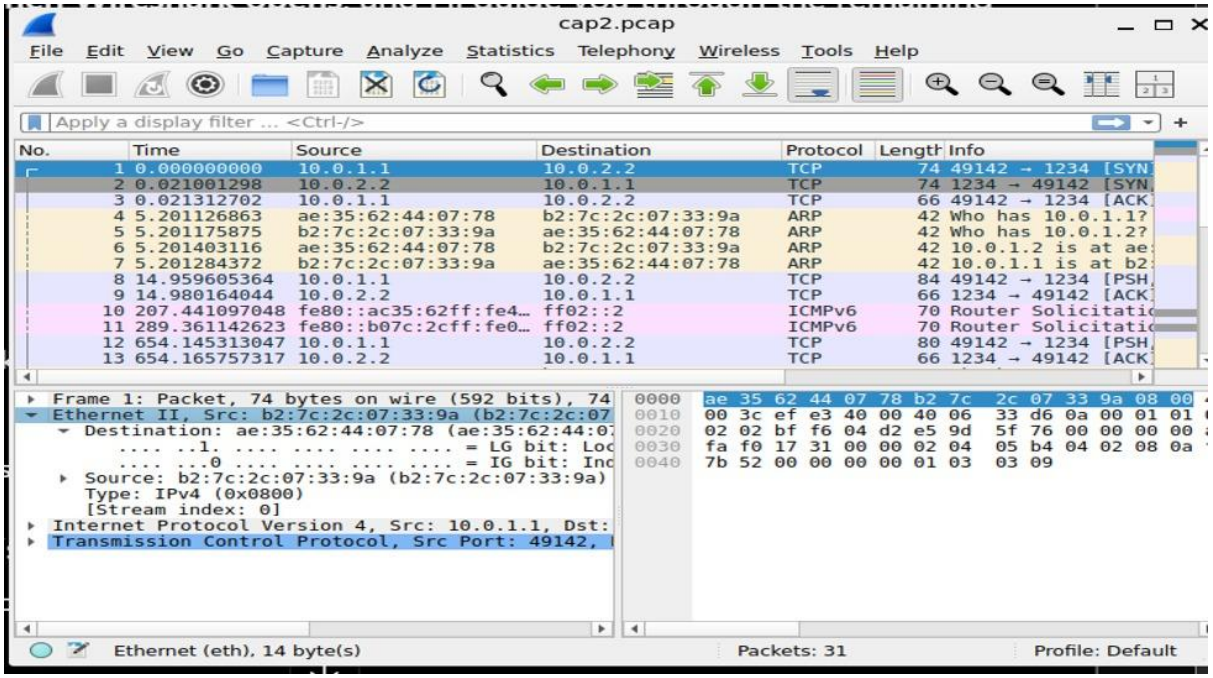
```
kavya_aluru@Kavya:~$ sudo ip netns list
host2
router
host1
kavya_aluru@Kavya:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
kavya_aluru@Kavya:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
kavya_aluru@Kavya:~$
```

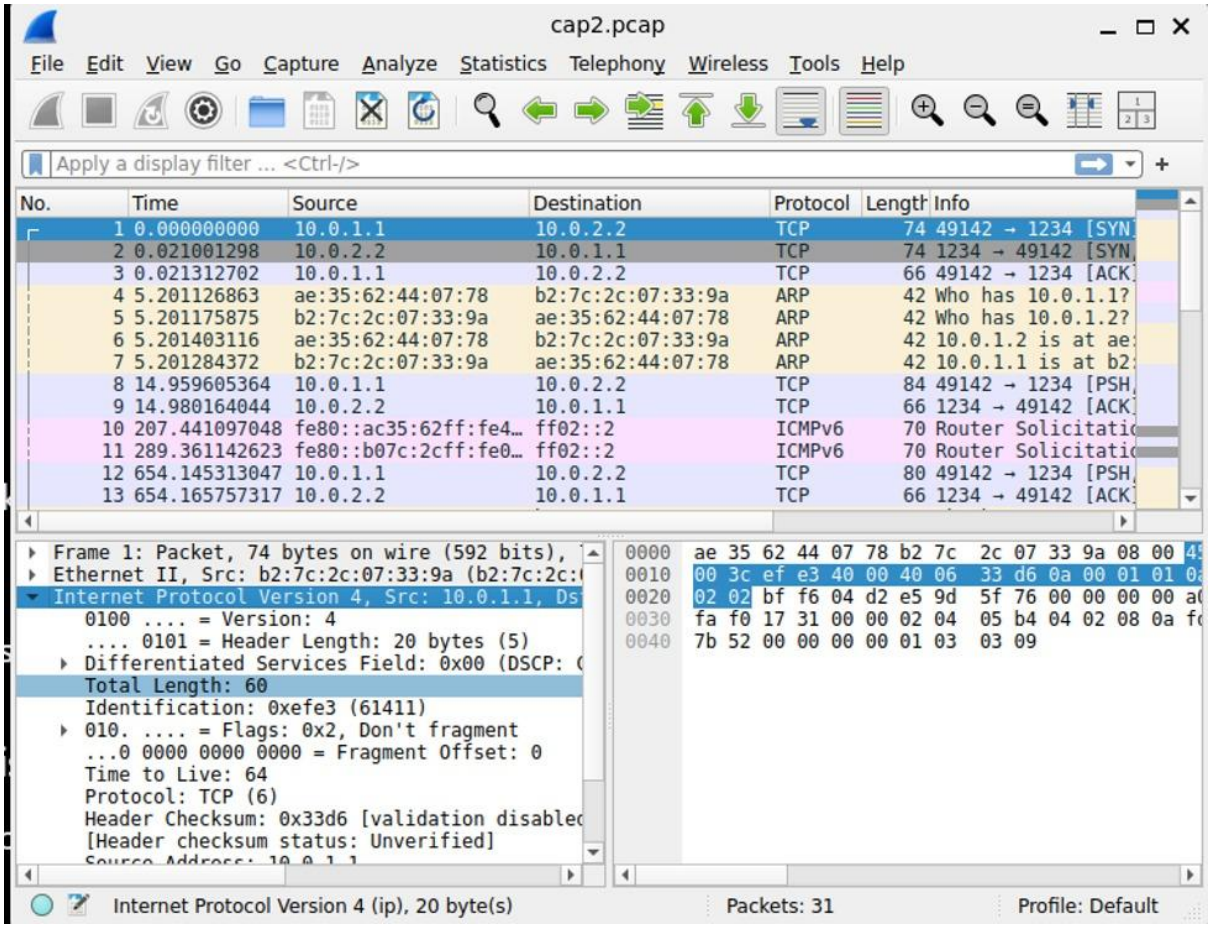
```
kavya_aluru@Kavya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev ve
th-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
kavya_aluru@Kavya:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev v
eth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
kavya_aluru@Kavya:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev ve
th-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
```

```
kavya_aluru@Kavya:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward
=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
kavya_aluru@Kavya:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.063 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.048 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.065 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2025ms
rtt min/avg/max/mdev = 0.048/0.058/0.065/0.007 ms
kavya_aluru@Kavya:~$
```




cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 TCP 74 49142 → 1234 [SYN]
2 0.021001298 10.0.2.2 10.0.1.1 TCP 74 1234 → 49142 [SYN,
3 0.021312702 10.0.1.1 10.0.2.2 TCP 66 49142 → 1234 [ACK]
4 5.201126863 ae:35:62:44:07:78 b2:7c:2c:07:33:9a ARP 42 Who has 10.0.1.1?
5 5.201175875 b2:7c:2c:07:33:9a ae:35:62:44:07:78 ARP 42 Who has 10.0.1.2?
6 5.201403116 ae:35:62:44:07:78 b2:7c:2c:07:33:9a ARP 42 10.0.1.2 is at ae:
7 5.201284372 b2:7c:2c:07:33:9a ae:35:62:44:07:78 ARP 42 10.0.1.1 is at b2:
8 14.959605364 10.0.1.1 10.0.2.2 TCP 84 49142 → 1234 [PSH,
9 14.980164044 10.0.2.2 10.0.1.1 TCP 66 1234 → 49142 [ACK]
10 207.441097048 fe80::ac35:62ff:fe4… ff02::2 ICMPv6 70 Router Solicitatio
11 289.361142623 fe80::b07c:2cff:fe0… ff02::2 ICMPv6 70 Router Solicitatio
12 654.145313047 10.0.1.1 10.0.2.2 TCP 80 49142 → 1234 [PSH,
13 654.165757317 10.0.2.2 10.0.1.1 TCP 66 1234 → 49142 [ACK]
Frame 1: Packet, 74 bytes on wire (592 bits), 74
Ethernet II, Src: b2:7c:2c:07:33:9a (b2:7c:2c:07
Destination: ae:35:62:44:07:78 (ae:35:62:44:07
.... ..1. .... = LG bit: Loc
....0 .... = IG bit: Inc
Source: b2:7c:2c:07:33:9a (b2:7c:2c:07:33:9a)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.0.1.1, Dst:
Transmission Control Protocol, Src Port: 49142,
0000 ae 35 62 44 07 78 b2 7c 2c 07 33 9a 08 00
0010 00 3c ef e3 40 00 40 06 33 d6 0a 00 01 01
0020 02 02 bf f6 04 d2 e5 9d 5f 76 00 00 00 00
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a
0040 7b 52 00 00 00 00 01 03 03 09
Ethernet (eth), 14 byte(s)
Packets: 31
Profile: Default


cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 TCP 74 49142 → 1234 [SYN]
2 0.021001298 10.0.2.2 10.0.1.1 TCP 74 1234 → 49142 [SYN,
3 0.021312702 10.0.1.1 10.0.2.2 TCP 66 49142 → 1234 [ACK]
4 5.201126863 ae:35:62:44:07:78 b2:7c:2c:07:33:9a ARP 42 Who has 10.0.1.1?
5 5.201175875 b2:7c:2c:07:33:9a ae:35:62:44:07:78 ARP 42 Who has 10.0.1.2?
6 5.201403116 ae:35:62:44:07:78 b2:7c:2c:07:33:9a ARP 42 10.0.1.2 is at ae:
7 5.201284372 b2:7c:2c:07:33:9a ae:35:62:44:07:78 ARP 42 10.0.1.1 is at b2:
8 14.959605364 10.0.1.1 10.0.2.2 TCP 84 49142 → 1234 [PSH,
9 14.980164044 10.0.2.2 10.0.1.1 TCP 66 1234 → 49142 [ACK]
10 207.441097048 fe80::ac35:62ff:fe4… ff02::2 ICMPv6 70 Router Solicitatio
11 289.361142623 fe80::b07c:2cff:fe0… ff02::2 ICMPv6 70 Router Solicitatio
12 654.145313047 10.0.1.1 10.0.2.2 TCP 80 49142 → 1234 [PSH,
13 654.165757317 10.0.2.2 10.0.1.1 TCP 66 1234 → 49142 [ACK]
Frame 1: Packet, 74 bytes on wire (592 bits),
Ethernet II, Src: b2:7c:2c:07:33:9a (b2:7c:2c:
Internet Protocol Version 4, Src: 10.0.1.1, Ds
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP:
Total Length: 60
Identification: 0xefe3 (61411)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x33d6 [validation disable
[Header checksum status: Unverified]
0000 ae 35 62 44 07 78 b2 7c 2c 07 33 9a 08 00
0010 00 3c ef e3 40 00 40 06 33 d6 0a 00 01 01
0020 02 02 bf f6 04 d2 e5 9d 5f 76 00 00 00 00
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a
0040 7b 52 00 00 00 00 01 03 03 09
Internet Protocol Version 4 (ip), 20 byte(s)
Packets: 31
Profile: Default